

Body temperature

It is the balance between the heat produced and the heat lost from the body, measured in degrees.



- Average normal oral temperature for adults is 37°C .
- Average normal rectal temperature is 37.5°C .
- Average normal axillary temperature is 36.5°C .
- Normal range of body temperature is $36.6 - 37.2^{\circ}\text{C}$.

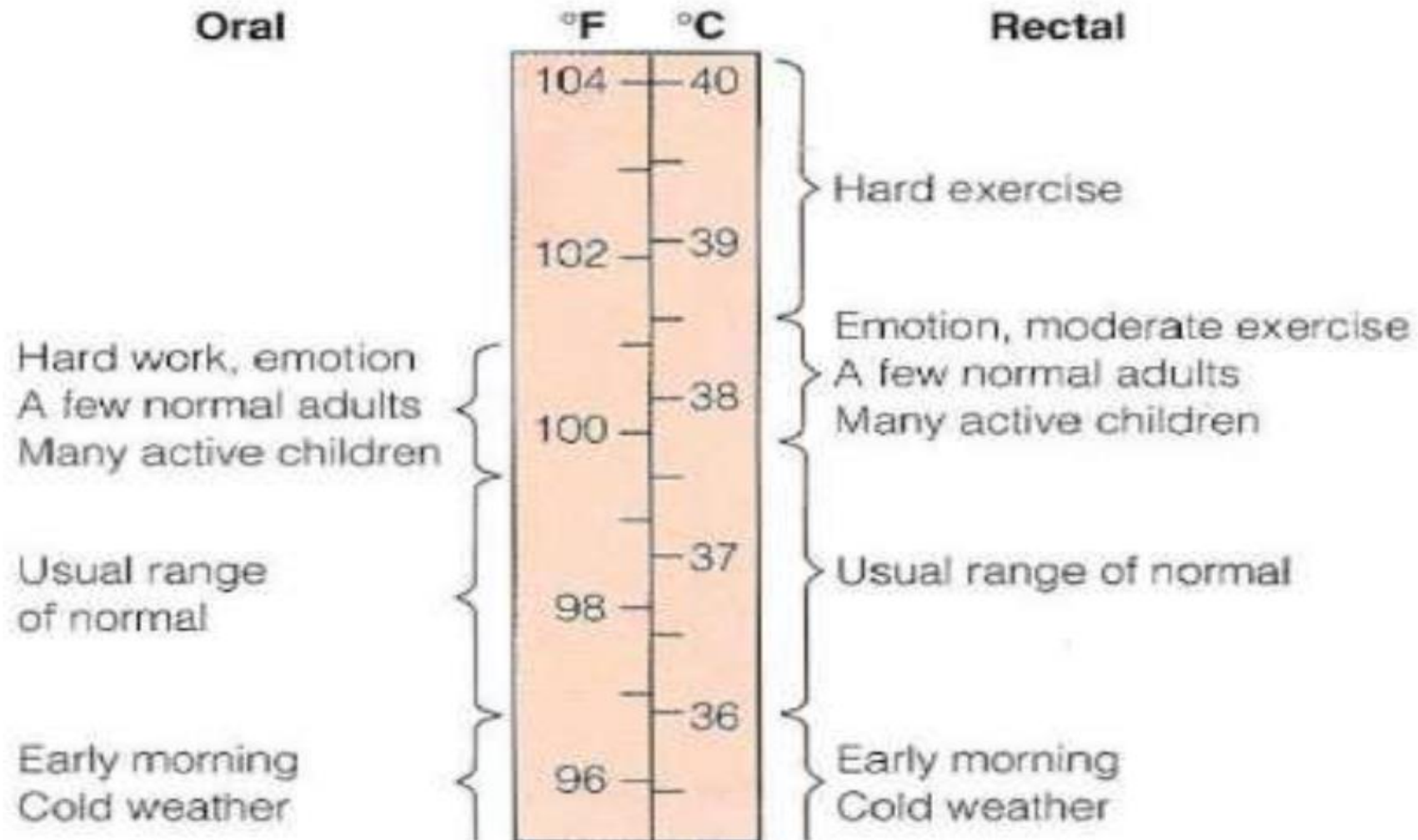
There are 2 kinds of body temperature; Core and surface temperature.

- Core temperature is the temperature of the deep tissues of the body, as the abdominal cavity and pelvic cavity. It remains relatively constant.
- Surface temperature is the temperature of the skin, the subcutaneous tissue, and fat. It, by contrast, rises and falls in response to the environment.

Routes for measuring body temperature:

- Oral: the thermometer is placed in the mouth.
- Axillary: obtained from the axilla.
- Rectal: from the rectum.
- Tympanic: from the ear canal

Estimated ranges of body temperatures in normal person



Temperature regulation:

- ☐ Physical control
- ☐ Chemical control
- ☐ Nervous system control

Physical control

- The body gains heat from its environment, as clothing, sun and ingestion of hot food



Chemical control

- The body produces heat through the metabolism of food. Body metabolism increases in order to produce more heat for body as necessary.
- An increase in the metabolic rate will result in an increase in body temperature and vice versa.

Nervous system control

- Body temperature is maintained by the hypothalamus in the central nervous system, located at the base of the brain.

The anterior portion of the hypothalamus is concerned with heat dissipation (loss),

- The posterior portion of the hypothalamus governs heat conservation (gain).

Factors increasing heat production

- 1- Muscle activity, as shivering, increases the metabolic rate.
- 2- Thyroxine output, increased thyroxine output increases the rate of cellular metabolism throughout the body.

This effect is called chemical thermogenesis, the stimulation of heat production in the body through increased cellular metabolism.

- 3- Basal metabolic rate (BMR), is the rate of energy utilization in the body required to maintain essential activities such as breathing.
4. Fever, increases the cellular metabolic rate and thus increases the body's temperature further.

5. Infections, increases the body's temperature

6. Emotion, stimulate the sympathetic nervous system with the release of epinephrine and norepinephrine, which increases the metabolic activities of body tissues which in turn increases heat production.

Mechanism of heat loss

Body heat is lost through

- 1- Skin,
- 2- Lungs, and
- 3- Excretion (digestive and urinary tracts).

The loss through the skin accounts about 85% of the total.



I. Radiation

It is the transfer of heat from the surface of one object to the surface of another without contact between the two objects

mostly in the form of infrared rays.



II - Convection

- Heat is lost when air currents pass over a warm object.

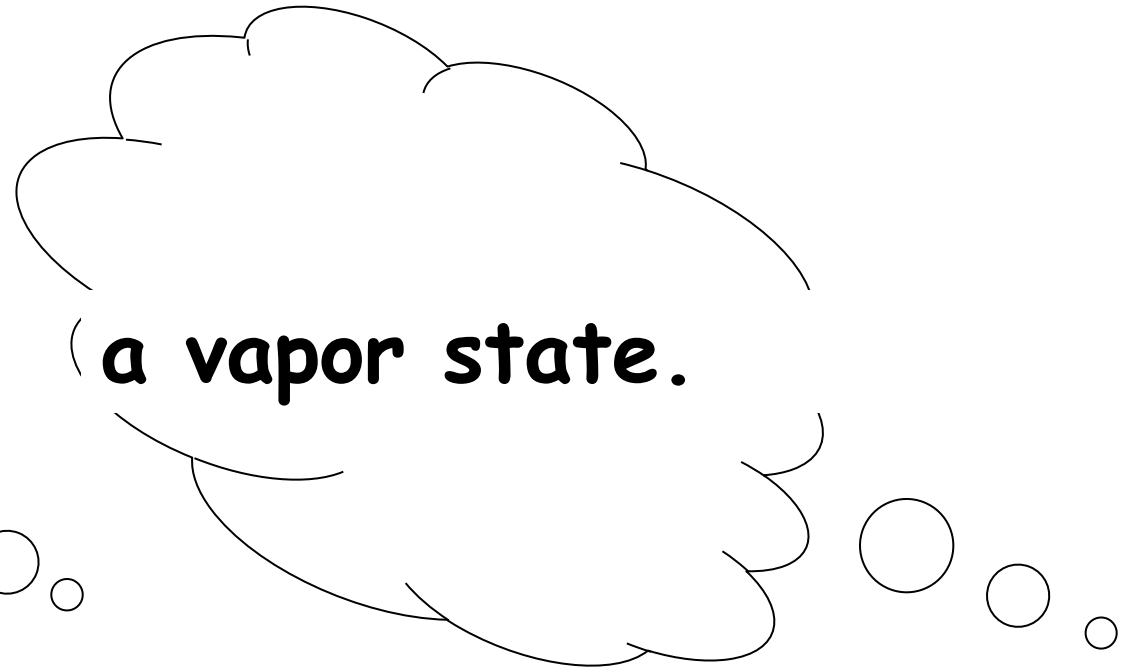
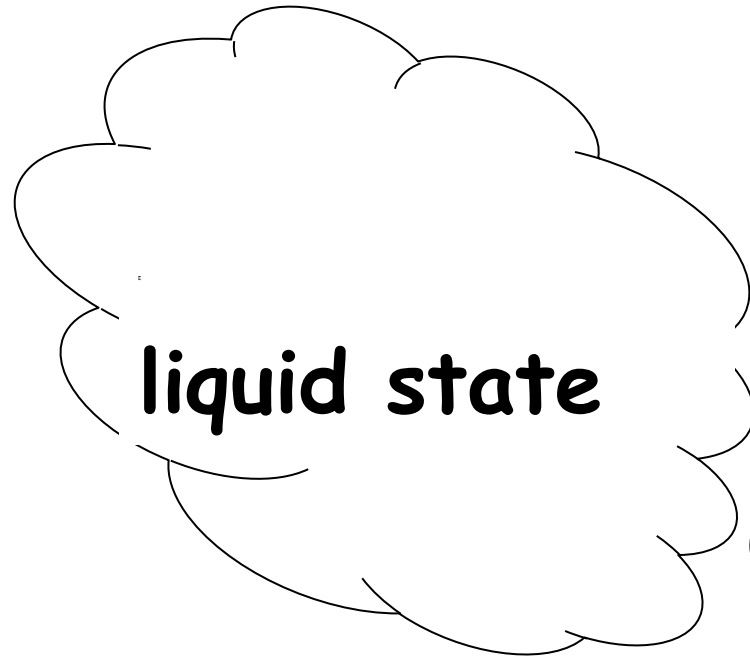
- Using of fan, open doors and windows to create drafts.



III - Evaporation



-It is the process whereby a substance is changed to:



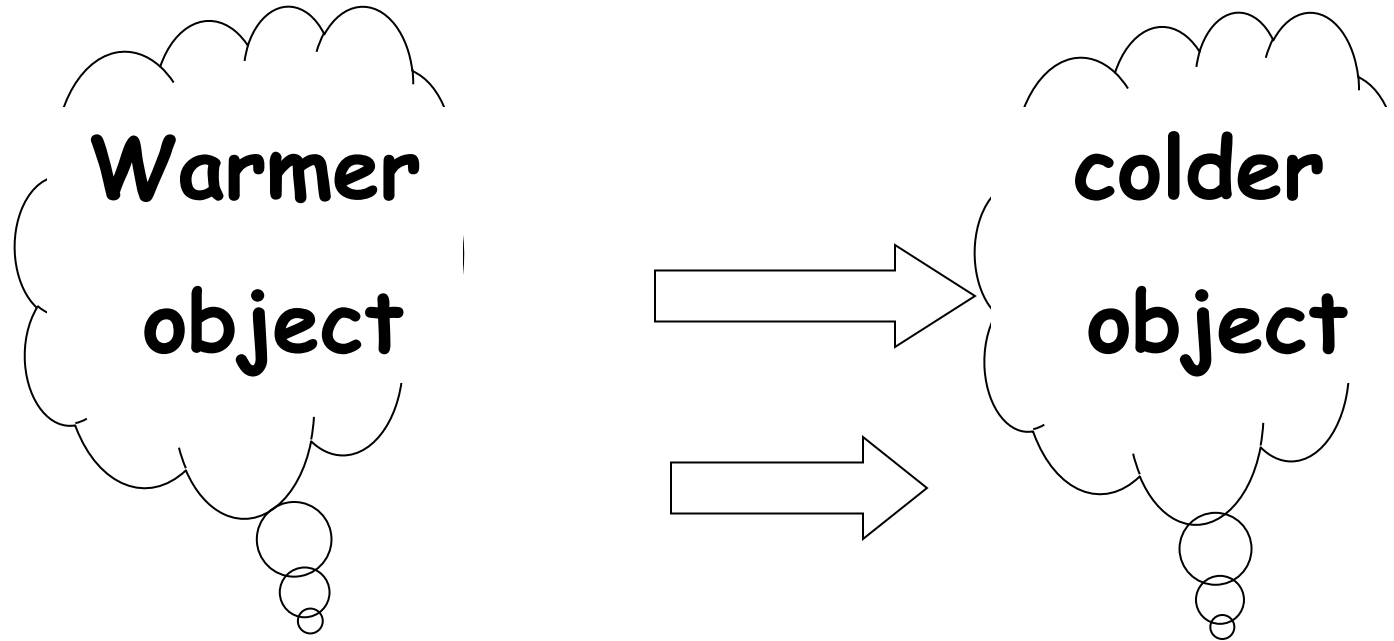
As sponging a patient with alcohol and water provide for cooling by evaporation



IV - Conduction

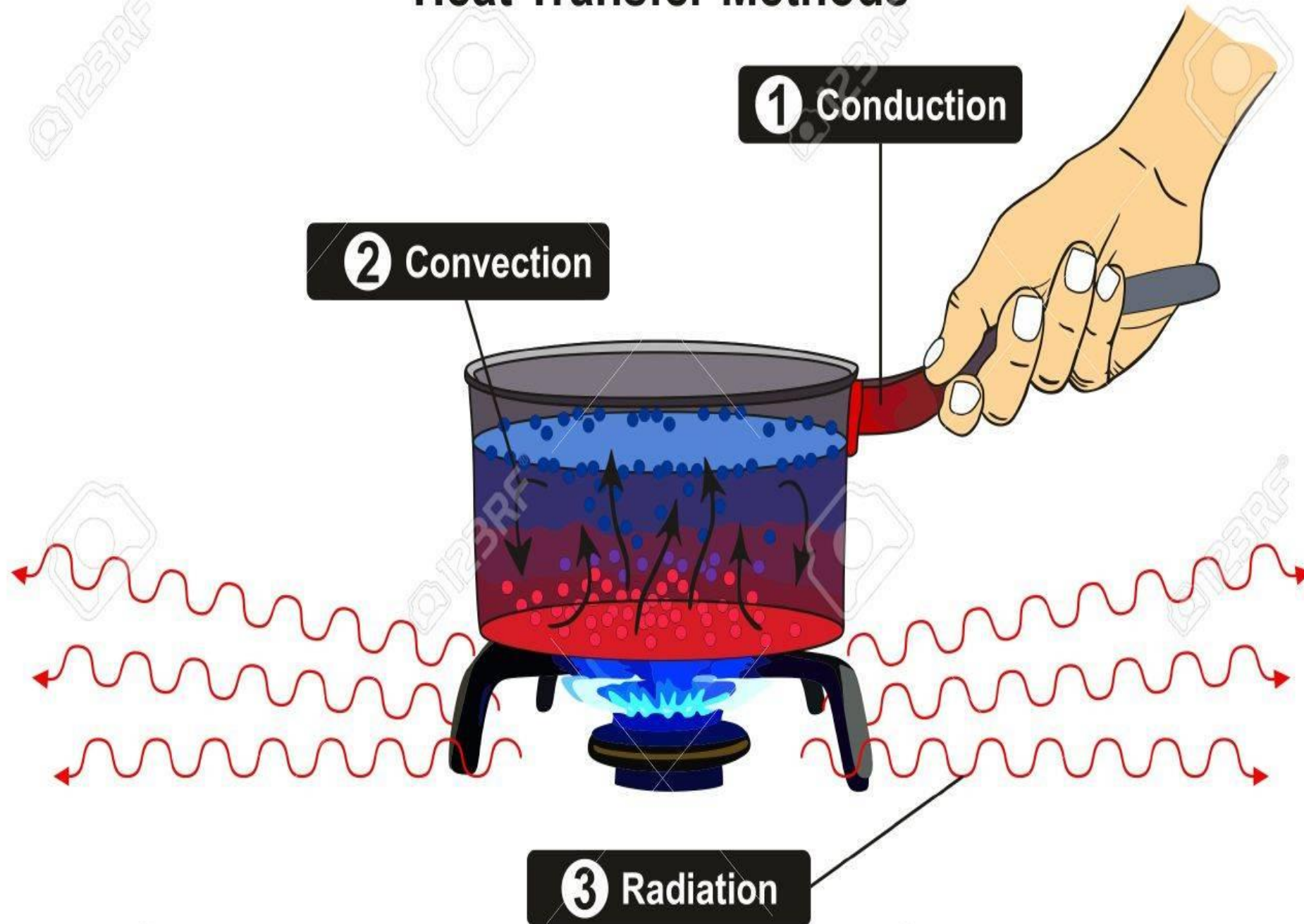


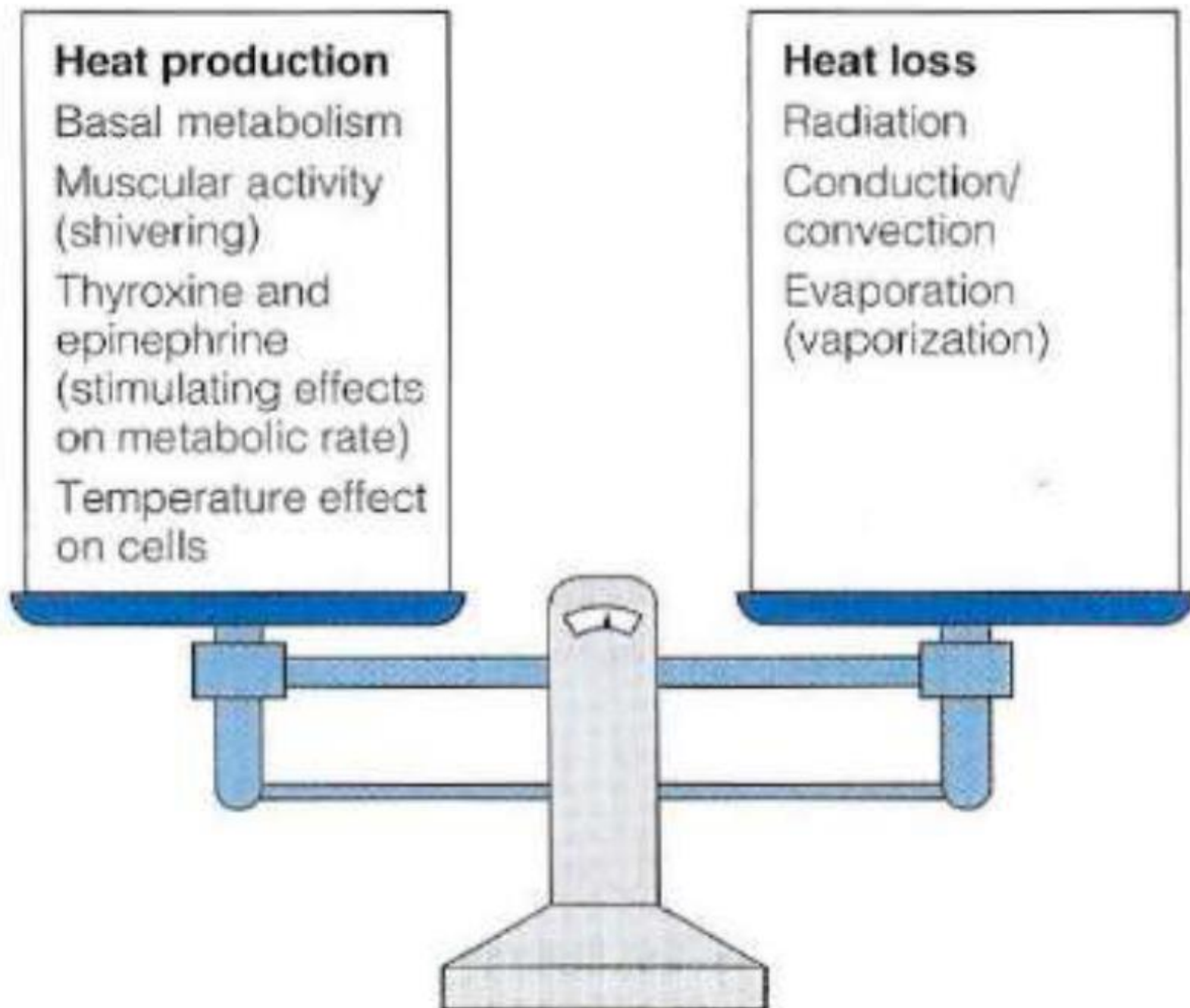
- It is the transfer of heat by direct contact between two objects.



As when a body is immersed in cold water.

Heat Transfer Methods





Factors affecting body temperature

- 1- **Age.** Children's temperatures continue to be more variable than those of adults until puberty
- 2- **Diurnal** variations Body temperatures normally change throughout the day
- 3- **Exercise.** increase body temperature.
- 4- **Sleep.** body temperature is lowered

5- Hormones. progesterone secretion at the time of ovulation raises body temperature by about 0.3°C to 0.6°C above basal temperature.

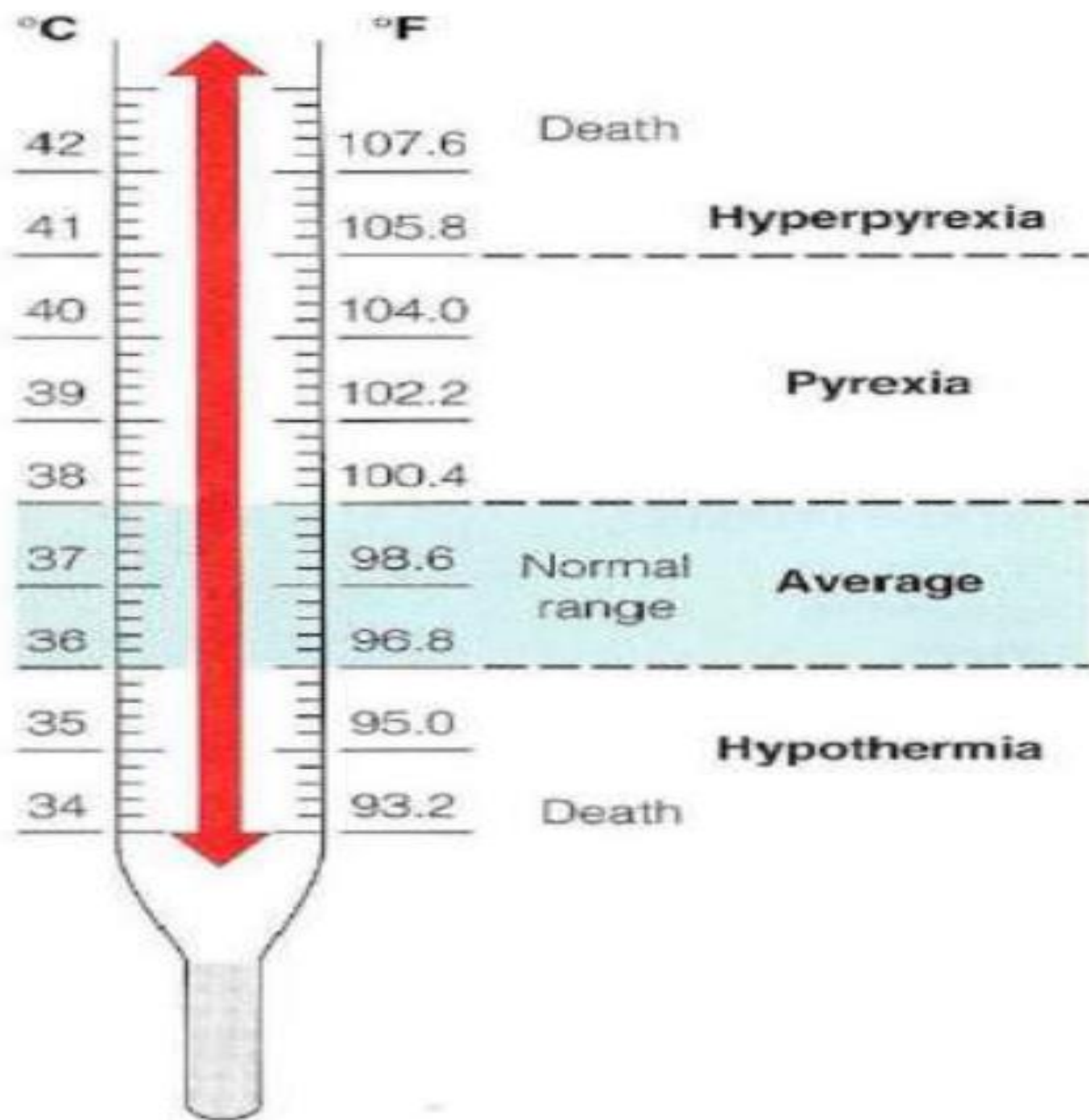
6-Stress. Stimulation of the sympathetic nervous system can increase the production of epinephrine and norepinephrine, thereby increasing metabolic activity and heat production.

7- Environment. environmental temperatures can affect a person's temperature regulation system.

• Alterations in body temperature:

There are 2 primary alterations in body temperature: pyrexia and hypothermia.

- A body temperature above the usual range is called pyrexia, hyperthermia, or fever.
- A very high fever, such as 41 °C (105.8°F), is called hyperpyrexia.
- The person who has a fever is referred to as febrile; the one who does not is afebrile.



Cause of fever

1- Damage of heat-regulating center due to head injuries or Cerebrovascular accidents and abnormally high body temperature.

This leads to disturbance in the heat regulating mechanism.

2- Acute infectious disease e.g. Malaria.

3- Acute inflammatory conditions.

4- Acute and prolonged pain.

5- Extreme nervousness.

6- Trauma or injury to body tissues.

Types of fever

1-Intermittent fever: the body temperature alternates at regular intervals between periods of fever and periods of normal or subnormal temperatures.

2-A remittent fever: a wide range of temperature fluctuations (more than 2°C) occurs over the 24-hour period.

3- A relapsing fever, short febrile periods of a few days are interspersed with periods of 1-2 days of normal temperature.

4- A constant fever, the body temperature fluctuates minimally but always remains above normal.

- Signs and symptoms of fever

Onset (Cold or chill phase)

- Increased heart rate and respiratory rate.
- Shivering, cold skin
- Cyanotic nail beds
- A "Gooseflesh" appearance of the skin
- Cessation of sweating

Course (Plateau phase)

- Absence of chills , skin that feels warm
- Photosensitivity , increased pulse and respiratory rates
- Increased thirst , mild to severe dehydration
- Drowsiness, restlessness, delirium, or convulsions

Defervescence / flush phase)

- Skin that appears flushed and feels warm
- Sweating
- Decreased shivering
- Possible dehydration

Measures to reduce body temperature

- Monitor vital signs.
- Assess skin color.
- Monitor white blood cell count, for indications of infection.
- Remove excess blankets when the client feels warm, but provide extra warmth when the client feels chilled.

Caring to those with fever:

- Provide adequate nutrition and fluids (e.g., 2,500–3,000 ml per day) to meet the increased metabolic demands and prevent dehydration.
- Measure intake and output.
- Reduce physical activity to limit heat production, especially during the flush stage.

- Provide oral hygiene to keep mouth moist
- Provide a tepid sponge bath to increase heat loss through conduction.
- Provide dry clothing and bed linens.
- Administer antipyretics (drugs that reduce the level of fever) as ordered.

Pulse & Blood Pressure

Pulse :

- it is an alternate expansion and recoil of an artery as the wave of blood is forced through it by the contraction of the left ventricle.

It is measured from an artery by fingers as from

- Carotid (neck)
- Brachial (In front of elbow)
- Radial from wrist)

Or

- **Apical (from apex of heart)**

Blood Pressure:

- It is the force or amount of pressure exerted by the blood against walls of the arteries when the left ventricle contracts.
- It is measured by sphygmomanometer and stethoscope

Factors controlling blood pressure:

- Pumping action of the heart (myocardial strength).
- Peripheral vascular resistance.
- Elasticity of the blood vessels.
- Blood volume.
- Viscosity of blood.

Normal range:

Pulse: 60-90 beat/minute.

Blood pressure: $\frac{120 + 20}{80 + 15}$ mmHg.

Abnormalities in pulse:

- Tachycardia: rate is above 100 b/m or more.
- Bradycardia: rate is below 60 b/m.
- Arrhythmias: an irregular rhythm (pulse rhythm means that the heart beats are identical in force and separated by equal intervals).

Abnormalities in Blood pressure:

- **Hypertension:** Abnormal elevation of blood pressure than normal.
- **Hypotension:** Abnormal decrease in blood pressure below normal.

Postural (orthostatic) hypotension: :

- Is a low blood pressure associated with weakness or fainting due to sudden change in position as when rising to an erect position.

Factors affecting pulse and blood pressure:

- Age.
- Sex.
- Body built.
- Exercise.
- Emotion.
- Fever.
- Posture.
- Drugs.
- Obesity.
- Blood volume.

Respiration:

Is the gas exchange between
the individual and the environmental air
in which the individual
takes in
oxygen and
eliminates
carbon dioxide and water vapor.

Respiratory regulation (control):

- **Nervous control:** through the respiratory center in the medulla.
- **Chemical stimulation:** by the carbon dioxide level in the blood.

- **Normal respiratory rate:**
14-20 cycle/minute.

Abnormalities:

- Tachypnea: respiratory rate is above the normal level.
- Bradypnea: respiratory rate is below the normal level.

Factors affecting respiration:

- Age.
- Sex.
- Exercises.
- During digestion.
- Emotion.
- Pain & discomfort.
- Drugs.
- Heat & cold application.
- Hemorrhage & shock.
- Fever.
- Illness & infection.
- Toxins.